SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

*I **Lokesh S (192511037)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Question 1: Hospital Shift Scheduling using Backtracking Search

Problem Statement

A hospital needs to assign **3 doctors (D1, D2, D3)** to **3 shifts (Morning, Afternoon, Night)** while satisfying the following constraints:

- D1 cannot work Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor must be assigned exactly one shift.

Variables

- D1
- D2
- D3

Domain

Each doctor can be assigned one of the following shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 ≠ Night
2. D2 < D3
3. One doctor per shift
4. D3 ≠ Morning
5. Every doctor gets exactly one shift

Backtracking Search

Step 1

Assign

D1 → Morning

Constraint Check

✓ D1 is not Night.

Current Assignment

Doctor	Shift
D1	Morning

Step 2

Assign

D2 → Afternoon

Constraint Check

✓ Unique shift

Current Assignment

Doctor	Shift
D1	Morning
D2	Afternoon

Step 3
Assign
D3 → Night
Constraint Check
✓ D3 is not Morning
✓ D2 before D3
✓ One doctor per shift
✓ All doctors assigned

Final Valid Schedule

Doctor	Assigned Shift
D1	Morning
D2	Afternoon
D3	Night

Search Tree

```
        Start
          |
     D1 = Morning
          |
    D2 = Afternoon
          |
     D3 = Night
          |
    Goal State Found
```

Conclusion
The Backtracking Search successfully finds a valid assignment that satisfies all constraints without requiring backtracking.

Question 2: Robot Navigation using BFS

Problem Statement
A robot operates in a **5 × 5 grid** and must move from **Start (S)** to **Goal (G)**.
Constraints:

- Move Up, Down, Left, Right
- Cost of each move = 1
- Cannot cross obstacles
- Manhattan Distance heuristic is provided
- Determine the optimal path using BFS.

Assumed Grid

```
S . . X .

. X . X .

. . . . .
```

X X . X .

. . . . G

Legend:

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Manhattan Distance Formula

$$h(n) = | x_1 - x_2 | + | y_1 - y_2 |$$

Example
Start = (0,0)
Goal = (4,4)

$$h = | 4 - 0 | + | 4 - 0 | = 8$$

BFS Node Expansion

Step	Expanded Node	Queue
1	S	(1,0),(0,1)
2	(1,0)	(0,1),(2,0)
3	(0,1)	(2,0),(0,2)
4	(2,0)	(0,2),(2,1)
5	(0,2)	(2,1),(1,2)
6	(2,1)	(1,2),(2,2)
7	(1,2)	(2,2)
8	(2,2)	(2,3),(3,2)
9	(2,3)	(3,2),(2,4)
10	(3,2)	(2,4),(4,2)
11	(2,4)	(4,2),(3,4)
12	(4,2)	(3,4),(4,3)
13	(3,4)	(4,3),(4,4)
14	Goal Found	Stop

Optimal Path
S

↓

(1,0)

↓

(2,0)

↓

(2,1)

↓

(2,2)

↓

(2,3)

↓

(2,4)

↓

(3,4)

↓

G

Total Cost
Each move costs 1.
Number of moves = **8**
Total Cost = 8

Conclusion
Breadth-First Search explores nodes level by level and guarantees the shortest path when all move costs are equal.

Question 3: Autonomous Rescue Robot

a) Constraint Analysis
The robot operates in a partially observable disaster environment with limited information. The constraints influence its decisions as follows:

Constraint	Effect on Decision-Making
Up, Down, Left, Right movement	Restricts possible actions
Obstacles	Robot must avoid blocked paths
Limited sensing	Robot only knows adjacent cells, requiring continuous exploration
Movement cost = 1	Encourages shorter routes
Risk zone cost = +2	Robot prefers safer paths even if slightly longer

Constraint	**Effect on Decision-Making**
Dynamic knowledge	Robot updates its map after every move

b) Solution Approach
Chosen Strategy: Uniform Cost Search (UCS) with Online Search
Steps

1. Start from the initial position (S).
2. Observe adjacent cells using sensors.
3. Add reachable cells to the priority queue.
4. Expand the node with the lowest total cost.
5. Update the map whenever new obstacles or risky zones are detected.
6. Repeat until the survivor (G) is reached.

Algorithm
Start

↓

Observe neighbouring cells

↓

Update map

↓

Compute movement cost

↓

Select lowest-cost node

↓

Move

↓

Repeat until Goal

↓

Rescue Survivor

c) Application of AI Concepts
Intelligent Agent
The rescue robot is an intelligent agent because it perceives its environment through sensors and acts autonomously to achieve its goal.
Rationality

The robot always chooses the action that minimizes total travel cost while maximizing safety.

Environment Type

Property	Description
Observable	Partially Observable
Dynamic	Yes
Sequential	Yes
Stochastic	Yes
Single Agent	Yes

d) Justification

Uniform Cost Search is the most suitable algorithm because:

- It always finds the least-cost path.
- It considers both movement cost and additional risk cost.
- It adapts well to changing information during navigation.
- It guarantees an optimal solution when costs are non-negative.
- It supports safe and efficient rescue operations.

Conclusion

The rescue robot functions as an intelligent, rational agent operating in a partially observable and dynamic environment. By combining **Uniform Cost Search** with **online learning**, it efficiently reaches the survivor while minimizing cost and avoiding dangerous areas.